# Hardik Choudhary

Pittsburgh, PA | linkedin.com/in/hardik-choudhary3101 | Portfolio: hardik3101.github.io/portfolio

## EDUCATION

**Carnegie Mellon University** — Pittsburgh, PA
*Master of Science in Mechanical Engineering – Research* — May 2027
**Coursework:** Modern Control Theory, Nonlinear Control, Introduction to Deep Learning, Machine Learning and AI for Engineers

**Sardar Vallabhbhai National Institute of Technology (SVNIT)** — Surat, India
*Bachelor of Technology in Mechanical Engineering* — May 2024

## SKILLS

**Control and Planning:** Trajectory optimization, graph search (INSAT, A*), LQR, adaptive control, PID, state observers, EKF SLAM
**Robotics and Simulation:** MuJoCo, Drake, Isaac Sim, Pinocchio, SOFA, Webots, ROS
**Programming and Machine Learning:** Python, C++, MATLAB, Simulink, PyTorch, NumPy, pandas
**Hardware:** Arduino, ESP32, NVIDIA Jetson Nano, Raspberry Pi, Quanser QUBE-Servo 2
**Mechanical Design:** SolidWorks (PDM/PLM), ANSYS (FEA), CFD, GD&T, design optimization, CNC machining, 3D printing

## RESEARCH EXPERIENCE

**Carnegie Mellon University** — Pittsburgh, PA
*Graduate Student Researcher, On-Orbit Manipulation (Advisors: Howie Choset, Matt Travers)* — Aug 2025 – Present

- Extend the INSAT planner (interleaved graph search and trajectory optimization) to tumbling satellites with a space-time visibility graph, raising rendezvous success from **80% to 99%** over 200 randomized trials
- Develop grasp, base, and arm controllers that stop a satellite's spin within actuation limits in Drake, MuJoCo, and Isaac Sim
- Analyze grasps for Snaker, a snake robot that wraps a satellite's engine nozzle in a helical cage, showing it tolerates **up to 5x** more misalignment than a parallel-jaw gripper and resists spin about every axis
- Design a **1 m, 18-link** tendon-driven Snaker that coils into a helix with the minimum possible number of motors, since power and heat dissipation are limited in orbit, and build an early 3D-printed prototype

## ACADEMIC PROJECTS

**Deep Learning for Engine Remaining-Useful-Life Prediction** | *CMU* — Jan 2026 – May 2026

- Built the data pipeline, LSTM baseline, and metrics for a two-person PyTorch benchmark of LSTM, Transformer, and TCN models on NASA C-MAPSS turbofan data; the Transformer led at **13.66 ± 0.47** cycles RMSE over 3 seeds

**Engine-Powered Drone** | *SVNIT* — Jan 2024 – Jun 2024

- Designed a single-engine drone for long flights in remote areas, lighter and cheaper than one carrying large battery packs, with a contact-free magnetic-gear drive to all four propellers
- Controlled each propeller's speed by its gear gap through a servo-driven linkage; mapped speed against gap with an encoder and Arduino (up to **75%** of the ideal 6,000 RPM) for feedback control that improved flight stability

**RAKSH Autonomous Agriculture Robot** | *Team Lead, SVNIT* — Feb 2023 – Dec 2023

- Developed row-following navigation for polyhouses with sensor fusion, VSLAM, and vision-based crop-row detection, running ROS and Python across Jetson Nano, Raspberry Pi, and ESP32 boards
- Designed a slide-in 10-chemical spraying module, each container with a servo-driven valve on a shared line, so the robot sprays only the plants its machine-learning model flags as diseased
- Designed a lead-screw linear actuator that lifts the camera to leaf height and implemented a rocker-bogie suspension
- Led a team of **5** to the **Rs 1,00,000** Proof of Concept prize at Robofest-Gujarat 3.0

**Formula Student Electric Car** | *Chassis Head, Team Phoenix Racing, SVNIT* — Nov 2021 – Jan 2023

- Led the chassis department of a **30-member** team, cutting frame mass with ANSYS FEA within stiffness targets
- Calculated suspension hardpoints and designed the aero package of a car that won Concept Goals Management at **Formula Bharat 2023**

## PROFESSIONAL EXPERIENCE

**Indian Institute of Technology Gandhinagar** — Gandhinagar, India
*Project Assistant, Human-Centered Robotics Lab* — Jan 2025 – Jun 2025

- Designed a prosthetic foot whose forefoot tilts side to side like a human foot, over three SolidWorks iterations from a single piece to a four-part assembly fitting a standard pylon mount, and built a test bed for repeatable loading
- Refined each design with ANSYS FEA, cutting the forefoot slits from four to two and comparing carbon fiber, POM, and PPS for energy storage and heel-strike impact